

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1163

Tait Electric Generating Station 100 kV
June 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1163. The Interconnection Customer proposes a 58.1 MW solar photovoltaic facility interconnecting at the Tait Electric Generating Station 100 kV substation in the Ohio area. The request was received on March 08, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 404 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,207,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tony Lappos-San Ardo 100kV	Tait Electric Generating Sta Substation-Hatch 100kV	110.5	151	42.2
San Ardo-Williams St 100kV	Tait Electric Generating Station-Hatch 100kV	108.6	153	42.2
Overlook-Hatch 100kV Line	Miami-Caro 100kV	108.5	126	32.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tony Lappos-San Ardo 100kV	Rebuild line with higher-capacity conductor	\$2,027,000	35% (\$718,000)
San Ardo-Williams St 100kV Line	Reconductor line	\$527,000	35% (\$187,000)
Hatch Substation-Overlook 100kV	Reconductor line	\$629,000	48% (\$302,000)
	Total allocated to project		\$1,207,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.